# COIT12209

## LECTURE NOTES

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# 1 Introduction

## 1.1 Electricity:

Through this is a steady of moving charges and their interaction until magnetic and electric fields change this. Charges move because a net force have been applied. The force comes from a difference in charge between two points on a circuit.

## 1.2 SI Units:

$$1\ Coulomb = 6.25 \times 10^{18} e^{-}$$

$$1A = \frac{1Q(Coulomb)}{1t(Seconds)}$$

## 1.3 Electrical Potential Energy:

The potential energy is closely associated with an object's position. $E_{PE}$ is what all electrical circuits have. To move charges within an electric field it requires an amount of work.

## 1.4 Voltage:

$$1V = \frac{1J}{C}$$

Defines the potential difference between two points in a conductor.

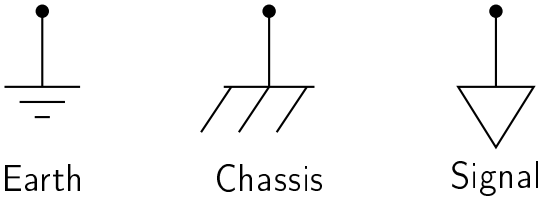
## 1.5 Elctro Motive Force:

emf is the energy provided by an eneyr source to enable the flow of charges.

## 1.6 Grounding:

When grounding a circuit we are basically joining it a giant resvior of both positive and negative charge. This is large enough that it can both recieve and donate charge. As such when an object comes into contact with group all net charge is removed becoming something with OV.

Ground Circuit Examples




## 1.7 Passive Circuit Element:

Conductors are circuit elements that allow the free movement of charge. Super conductors do this with no resistance. Insulators are the opposite of conductors. They prevent the free movement of charge. A semi-conductor has properties of conductors and insulators.

## 1.8 Resistance:

Resistivity ($\rho$) is how strongly it resists electrical current.

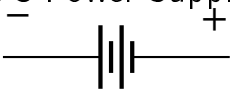
$$R = \frac{\rho L}{A}$$

Where:

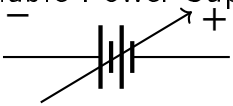
- R: Resistance ($\Omega$)
- $\rho$: Resistivity ($(\Omega \cdot m)^{-1}$)
- L: Lenght of wire ($m$)
- A: CSA ($m^2$)

### 1.9 Circuit Symbols:

DC Power Supply




Variable Power Supply




AC Power Supply



Inductor



Wire



Junction



Ampmeter



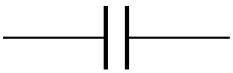

Volt meter




Resistor



Capacitor



Voltage rises across from the negative to positive terminals of the components. In addition to the opposite flow to the amps. Current divider rule:

$$I_1 = I_S \cdot \frac{R_2}{R_1 + R_2}$$

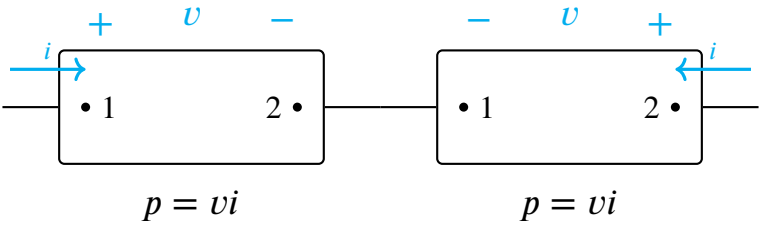
$$I_2 = I_S \cdot \frac{R_1}{R_1 + R_2}$$

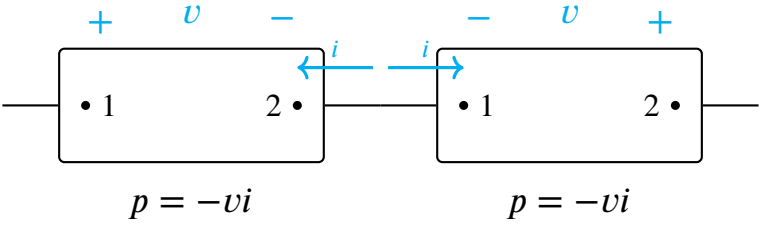
# 2 Circuit Simplification

## 2.1 Ideal Power Sources:

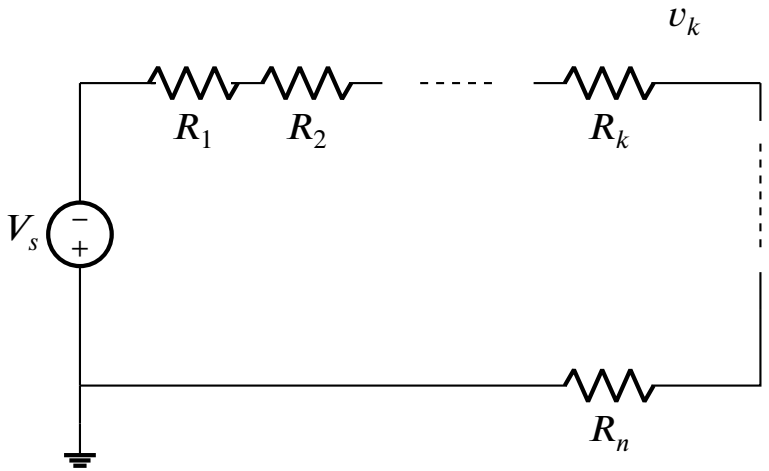
For ideal voltage and current sources, there are independent and dependent sources. The independent source are illustrated by a circle around the indicator and its numerical value is just defined. The dependent source usually rely on a contast multiplied by a either a current or voltage value within the circuit. They are also illustrated through a diamond shape.

## 2.2 Polarity and Power:



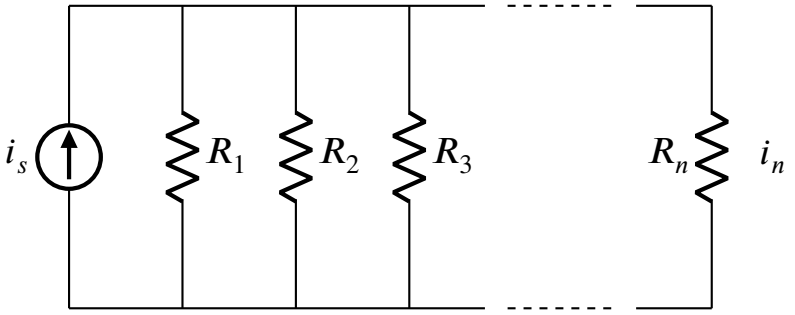
## 2.3 Voltage Divider Rule:




For this to work the current through all resistors must be equal. The highest voltage drop occurs over the largest resistor.

$$V_k = \frac{R_k}{R_{eq}} \cdot V_S$$

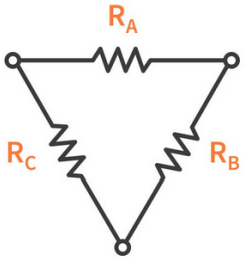
## 2.4 Current Divider Rule:

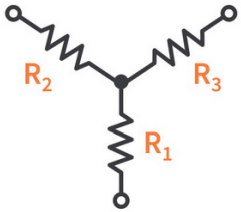



The highest current goes through the smallest resistor.

$$I_k = \frac{I_S}{R_k R_{eq}}$$

## 2.5 Delta-Wye Resistor Configurations:




$$R_a = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1}$$

$$R_b = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_2}$$

$$R_c = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_3}$$

$$R_1 = \frac{R_bR_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_cR_a}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_aR_b}{R_a + R_b + R_c}$$